Product Requirements Document (PRD)

Location Risk Advisor for First-Time Home Buyers

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1. Product Overview

Product Name

Location Risk Advisor for First-Time Home Buyers

Product Summary

Location Risk Advisor is a web-based decision-support platform designed to help first-time home buyers in Greater Jakarta (Jabodetabek) evaluate the risks and suitability of residential locations before purchasing a home.

Unlike traditional property listing platforms that primarily focus on price, property specifications, and promotional content, this product focuses on helping users understand the real-world risks and long-term livability of a location.

The platform provides simplified, actionable insights related to:

- Flood risk
- Daily mobility and commute burden
- Transportation accessibility
- Nearby facilities and services
- Environmental quality
- Road and physical access conditions
- Area development prospects
- Long-term residential suitability

The goal is to help users make more confident and informed housing decisions before committing to booking fees or down payments.

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2. Problem Statement

First-time home buyers often make high-stakes purchasing decisions without having sufficient visibility into the actual risks and quality of a residential area.

Most existing property platforms emphasize:

- Property pricing
- House photos
- Building specifications
- Developer promotions

However, they provide limited support in evaluating critical location-related risks such as:

- Flood exposure
- Traffic congestion
- Transportation access
- Environmental conditions
- Infrastructure quality
- Accessibility to daily facilities
- Long-term neighborhood suitability

As a result, buyers may unknowingly choose locations that negatively affect their daily lives, commute experience, family needs, or long-term quality of living.

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3. Product Goals

Primary Goal

Help first-time home buyers clearly understand and evaluate location risks before purchasing a property.

Key Objectives

1. Help users identify residential areas that match their lifestyle and budget
2. Help users compare multiple locations objectively
3. Reduce the risk of choosing unsuitable or high-risk locations
4. Simplify complex geographic and infrastructure data into understandable insights
5. Improve buyer confidence during the property evaluation process

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4. Target Users

Primary Audience

First-time home buyers in the Greater Jakarta metropolitan area (Jabodetabek) purchasing homes primarily for personal residence rather than investment purposes.

User Characteristics

- Working-age adults
- Stable income earners

• Limited home-buying experience
• Budget-conscious buyers
• Often considering suburban areas such as:
• Bekasi
• Depok
• Tangerang
• Bogor
• Other satellite cities

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5. User Personas

Persona 1 — Young Professionals or Couples

Needs

• Efficient commute
• Transportation access
• Affordable housing
• Access to daily lifestyle facilities

Primary Concern

> “I don’t want to buy a cheap house that creates exhausting daily commutes.”

Persona 2 — Newly Married Couples

Needs

• Long-term residential suitability
• Safe neighborhoods
• Family-friendly facilities
• Future child planning considerations

Primary Concern

> “Will this location still work well when we start a family?”

Persona 3 — Young Families with Children

Needs

• School access
• Healthcare facilities
• Flood safety
• Comfortable environment
• Daily convenience

Primary Concern

> “Is this area safe and comfortable for raising children?”

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6. Core User Insight

Across all user segments, the primary shared concern is:

> Users are afraid of making the wrong housing decision because they do not fully understand location-related risks.

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7. Product Value Proposition

For Users

The platform helps buyers make more informed and confident decisions by translating complex location data into practical residential insights.

Differentiation

Unlike conventional property marketplaces, this product focuses on:

> **Location Risk Intelligence**

Rather than helping users simply find houses, the platform helps them evaluate whether a location is genuinely suitable for living.

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8. User Journey & Usage Context

Stage 1 — Early Exploration

User Questions

- “Which areas fit my budget and lifestyle?”
- “Which areas are relatively safe and practical?”

User Needs

- Area recommendations
- High-level risk maps
- Budget-based filtering
- Alternative area suggestions

Stage 2 — Comparing Candidate Locations

User Questions

- “Which location has the lowest acceptable risk?”
- “Which option best fits my daily routine?”

User Needs

- Side-by-side location comparison
- Risk scoring
- Commute estimation
- Facility comparison
- Pros and cons analysis

Stage 3 — Pre-Booking or Pre-DP Validation

User Questions

- “What hidden risks should I check before paying?”

User Needs

- Due diligence checklist
- Red flag warnings
- Survey guidance
- Questions for agents or residents
- Downloadable risk reports

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9. Core Risk Categories

The platform should evaluate locations across five primary categories:

1. Environmental Risk

Includes:

- Flood exposure
- Pollution
- Noise levels
- Population density

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2. Mobility Risk

Includes:

- Traffic congestion
- Commute burden
- Public transportation access
- Road accessibility

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3. Daily Facility Risk

Includes:

- Schools
- Hospitals
- Clinics
- Markets
- Supermarkets
- Family services

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4. Area Growth Risk

Includes:

- Infrastructure development
- Long-term growth potential
- Property appreciation indicators

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5. Physical Access Risk

Includes:

- Road width
- Road conditions
- Entry/exit bottlenecks
- Cluster access quality

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10. MVP Scope

Priority MVP Goal

The first release should primarily focus on:

Helping users avoid choosing unsuitable or high-risk residential locations.

MVP Features

1. Flood Risk Map

- Flood-prone area visualization
- Flood history indicators
- River proximity analysis

2. Location Suitability Score

Example:

- Residential suitability score
- Flood risk score
- Commute convenience score
- Family facility score

3. Multi-Location Comparison

Users can compare:

- 2–5 locations
- Risk categories
- Commute burden
- Facility accessibility

4. Commute Estimation

- Office-to-home travel estimation
- Transportation mode comparison
- Congestion indicators

5. Red Flag Alerts

Examples:

- Severe flood risk
- Poor transport access
- High congestion
- Weak infrastructure

6. Location Risk Report

Downloadable/shareable summary including:

- Overall risk score
- Key concerns
- Strengths
- Validation checklist
- Recommended follow-up questions

7. Survey Checklist

Field validation support for:

- Road conditions
- Flood indicators
- Environmental quality
- Traffic conditions
- Neighborhood observations

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11. Future Enhancements (Post-MVP)

Planned Features

- Personalized onboarding profiles
- Smart alternative area recommendations
- Saved shortlist management
- User field notes and photo uploads
- Interactive map layers
- Confidence scoring for data reliability
- Educational content for first-time buyers
- Collaborative sharing with family members
- AI-generated residential recommendations

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12. UX & Information Design Principles

Design Philosophy

Because the target users are non-experts, the platform should prioritize:

- Simplicity
- Clarity
- Decision-oriented language

Preferred Presentation Style

Instead of technical metrics:

- Flood Risk: High
- Commute Suitability: Moderate
- Family Facilities: Strong
- Public Transport Access: Weak

UX Principles

- Progressive disclosure

- Summary-first presentation
- Mobile-friendly design
- Map-first exploration
- Clear visual scoring

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13. Data & Integration Requirements

Potential Data Sources

- Flood and environmental data
- Public transportation data
- Traffic and commute APIs
- Government infrastructure plans
- Public facilities databases
- Commercial location datasets
- User-contributed observations

Technical Considerations

- Data quality validation
- Data confidence scoring
- Geographic processing
- Map rendering performance
- Mobile responsiveness

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14. Product Risks

1. Data Accuracy Risk

Risk

Incorrect or outdated data may reduce trust.

Mitigation

- Confidence score indicators
- Source transparency
- Clear disclaimers
- Encourage field validation

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2. Product Complexity Risk

Risk

Too much information may overwhelm users.

Mitigation

- Simplified scoring
- Layered information architecture
- Guided workflows
- Progressive disclosure

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3. Recommendation Bias Risk

Risk

Users may perceive favoritism toward certain developers or areas.

Mitigation

- Transparent scoring logic
- Balanced pros and cons
- Separation between data and recommendations

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4. Over-Reliance Risk

Risk

Users may depend entirely on the platform without conducting site visits.

Mitigation

- Strong survey recommendations
- Validation checklists
- Confidence indicators
- Resident interview prompts

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15. Success Metrics

User Metrics

- Number of location comparisons completed
- Risk reports generated
- Survey checklists used

- Shortlisted properties saved

Engagement Metrics

- Repeat usage during house-search process
- Session duration
- Number of areas evaluated per user

Product Metrics

- User trust score
- Recommendation satisfaction
- Reduction in abandoned property consideration after risk discovery

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16. Monetization Opportunities

Potential monetization models include:

- Premium location reports
- Subscription plans
- Partner integrations
- Mortgage and financing referrals
- Property platform partnerships
- Lead generation
- Sponsored infrastructure insights

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17. Open Questions

1. Which feature should define the MVP experience?
2. Should the primary flow begin with maps, address input, or property comparison?
3. How should location risk scoring be calculated?
4. Which datasets are realistically available and reliable?
5. Should the platform support landed houses only or apartments as well?
6. Will monetization focus on B2C subscriptions or B2B partnerships?
7. Should the product integrate directly with property listings?
8. How can neutrality be maintained when working with developers or agents?
9. What report format is easiest for users to understand?
10. What are the primary success metrics for MVP launch?

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18. Recommended Next Steps

Product Strategy

- Define MVP scope and scoring framework

- Prioritize highest-value datasets
- Validate user pain points through interviews

UX & Research

- Design initial map exploration flow
- Prototype location comparison experience
- Test simplified scoring visualization

Technical

- Evaluate map and geospatial stack
- Validate external data providers
- Design data confidence architecture

Business

- Explore partnership opportunities
- Validate willingness to pay
- Define long-term monetization strategy
